Doga Tekin

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AI researcher & engineer working on language models since 2021. Fine-tuned language models for finance, led the development of LLM-based agents for cybersecurity and support, and now training transformers for structured data.

Education

EPFL 2018 – 2021
Computer Science MSc | GPA: 5.85 / 6.00 *Lausanne, CH*
- Graduated **1st among 112 students** in the CS department.

Bilkent University 2014 – 2018
Electrical & Electronics Engineering BSc | GPA: 3.79 / 4.00 *Ankara, TR*
- Received a scholarship for ranking in the **top 0.1% in Turkey** on the entrance exam.

Work Experience

Oracle Labs & Oracle Database AI Research 2021 – Present
Principal Member of Technical Staff (*2026–Present*) *Zurich, CH*
- Leading the development of an LLM-based agent for customer support request analysis and escalation.
  - Identifies 5x more critical cases than the existing manual process while cutting down false positives.
  - Being integrated to process 1M+ support requests annually, saving time and effort for >3000 support associates.
- Training Relational Graph Transformers to generate embeddings for entity similarity search on structured data.
  - Outperforms tabular / relational foundation model baselines such as TabICL and KumoRFM on similarity tasks.

Senior Member of Technical Staff (*2023–2026*)
- Led the development of an LLM-based agent that analyzes the cybersecurity of Oracle Cloud tenancies.
  - Worked across the stack building custom tools, evaluation environments, and a user interface for the agent.
  - Found 100s of concrete vulnerabilities in production tenancies scanned by our system.
- Developed a transformer-based method that turns tables into graphs for use in Oracle Finance products.
  - Filed a patent application (pending) for this method of fine-tuning language models on tabular data.
- Mentored and supervised 20+ interns on NLP, Graph ML, LLM, RL, and cybersecurity topics.

Member of Technical Staff (*2021–2023*)
- Built a pipeline for natural text generation from knowledge graphs by fine-tuning a T5 language model.
- Improved Graph ML explainability features in Oracle's graph toolkit, PGX.
- Used feature engineering and AutoML to train accurate trajectory prediction models for a major Oracle customer.

Research Intern (*2021–2021*)
- Fine-tuned language models to build a state-of-the-art pipeline turning financial documents to knowledge graphs.

Accenture Labs 2020 – 2020
Research Intern *Dublin, IE*
- Implemented evolutionary hyperparameter optimization algorithms in Accenture's graph library AmpliGraph.

Publications

NeurIPS 2019 OptRL Workshop
Reinforcement Learning with Langevin Dynamics – K. Parameswaran, **D. Tekin**, P. Rolland, V. Cevher
- Implemented a novel exploration method for reinforcement learning algorithms using TensorFlow.
- Conducted extensive testing on public RL benchmarks to produce results for the paper.

Skills

- **ML & AI**: LLMs, Agents, NLP, Graph ML, Relational Deep Learning, Reinforcement Learning
- **Frameworks & Libraries**: PyTorch, Hugging Face, scikit-learn, NumPy, pandas, PySpark
- **Programming**: Proficient in Python. Good with SQL, familiar with JavaScript. Used Java and C++ before.

Personal Projects

I blog about topics like drawing with code and experimenting with JAX, and share tools and games at dogatekin.com.